

Advanced Communication Networks

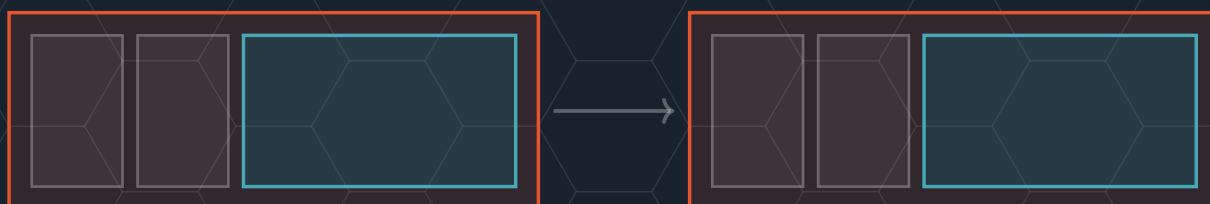
شبکه‌های ارتباطی پیشرفته

Module 13

S-GW (Serving Gateway) and P-GW (PDN Gateway)

S-GW (دروازه خدمات‌دهنده) و P-GW (دروازه PDN)

OUTER IP | UDP | GTP-U | USER IP PACKET



Lecture Notes

Instructor: Amin Azari

Compiled by Aidin Shekari

استاد: امین آذری github.com/AminAzari

In This Module

1	Why Separate S-GW and P-GW?	2
2	S-GW (Serving Gateway)	3
3	P-GW (PDN Gateway)	4
4	GTP Protocol	7
5	Packet Path Trace (End-to-End with Encapsulation)	9
6	CUPS: Control and User Plane Separation	12
7	Evolution: S-GW + P-GW → UPF (5G)	13
	Summary	15

1 Why Separate S-GW and P-GW?

LTE deliberately splits the user-plane gateway into two distinct nodes:

Role	S-GW	P-GW
Anchoring	Local mobility anchor (intra-LTE)	PDN/IP anchor (inter-RAT, global)
Scope	Per-MME pool area	Per-PDN connection (could span MME areas)
Mobility	Changes during inter-MME HO	Stays the same for entire PDN session
Placement	Close to RAN (low latency)	Centralized (policy, charging)

Design Rationale:

1. **Different lifetimes:** S-GW can relocate during mobility; P-GW remains fixed for IP continuity
2. **Independent scaling:** S-GW scales with local traffic; P-GW scales with subscribers/policies
3. **CUPS preparation:** The split enables Control/User Plane Separation — you can distribute S-GW-U at the edge while keeping S-GW-C centralized
4. **Roaming:** In home-routed roaming, S-GW is in visited PLMN; P-GW is in home PLMN (S8 interface)

2 S-GW (Serving Gateway)

2.1 User-Plane Forwarding

The S-GW is essentially a **GTP-U tunnel switch**:

- Receives DL packets from P-GW on S5/S8 GTP-U tunnel
- Forwards them to eNodeB (eNB) on S1-U GTP-U tunnel
- Receives UL packets from eNB on S1-U and forwards to P-GW on S5/S8

Each bearer has **separate GTP-U tunnels** on each interface (identified by TEID).

2.2 Local Mobility Anchor

During **X2 handover** (direct eNB-to-eNB):

- S-GW does NOT change
- Source eNB forwards buffered/in-transit packets to target eNB via X2
- After handover, MME sends **Modify Bearer Request** to S-GW with new eNB address/TEID
- S-GW switches S1-U tunnel endpoint from source eNB to target eNB (**path switch**)

This avoids tearing down the S5 tunnel to P-GW — only the S1-U leg changes.

2.3 Downlink Data Buffering

When UE is in **ECM-IDLE** (no S1-U tunnel exists):

1. DL packets arrive from P-GW on S5 tunnel
2. S-GW has no S1-U endpoint → **buffers packets**
3. S-GW sends **Downlink Data Notification** to MME (via S11)
4. MME initiates paging
5. After UE responds (Service Request), MME sends Modify Bearer Request with new eNB TEID
6. S-GW flushes buffered packets to eNB

Buffer limits: Configurable per operator; typically 1-10 packets per bearer. Excess dropped.

2.4 Per-UE/Per-Bearer Charging

S-GW generates **CDRs (Charging Data Records)** with:

- UL/DL volume per bearer
- Duration
- QoS information (QCI, ARP, GBR/MBR)
- Cause of record closure (HO, time limit, volume limit)

Sent to **offline charging system (CDF)** via Rf/Gz interface.

2.5 Lawful Interception

S-GW can mirror user-plane traffic for lawful interception (LI):

- X2 interface (ETSI standard) to Law Enforcement Monitoring Facility (LEMF)
- Triggered by warrant, activated via X1 interface from LI administration
- Duplicates packets without affecting UE service

2.6 S-GW Interfaces

Interface	Peer	Protocol	Function
S1-U	eNB	GTP-U (UDP)	User-plane to/from RAN
S5	P-GW (same PLMN)	GTP-U + GTPv2-C	UP forwarding + CP session mgmt
S8	P-GW (roaming)	GTP-U + GTPv2-C	Same as S5 but inter-PLMN
S11	MME	GTPv2-C	CP: bearer/session management
S12	UTRAN (3G)	GTP-U	Direct tunnel for 3G interworking

3 P-GW (PDN Gateway)

3.1 PDN Connectivity and IP Address Allocation

The P-GW is the **UE's gateway to the Internet/PDN** — it assigns the UE its IP address:

IPv4 Allocation Methods:

- **Static:** Pre-configured in HSS subscription data
- **Dynamic (DHCPv4):** P-GW acts as DHCP server or relays to external DHCP
- **Dynamic (internal pool):** P-GW allocates from local address pool (most common)

IPv6 Allocation:

- P-GW assigns a /64 prefix via **Router Advertisement (SLAAC)**
- UE generates Interface ID (EUI-64 or privacy extensions)
- Prefix delivered in PCO (Protocol Configuration Options) during bearer activation

Dual-Stack (IPv4v6):

- Both IPv4 address and IPv6 prefix assigned simultaneously
- Requires PDN Type = IPv4v6 in subscription

3.2 Policy Enforcement

P-GW enforces policies received from **PCRF** via the **Gx interface**:

- **PCC Rules (Policy and Charging Control):**
 - SDF (Service Data Flow) filters — 5-tuple matching
 - QoS parameters (QCI, GBR, MBR, ARP)
 - Charging keys and rating groups
 - Gate status (open/close specific flows)
- **TFT (Traffic Flow Template):**
 - **DL TFT:** At P-GW — maps incoming DL packets to correct bearer
 - **UL TFT:** Sent to UE — UE maps UL packets to correct bearer
 - Packet filters: source/dest IP, port range, protocol, ToS/DSCP

3.3 Charging

Type	Interface	Node	Use Case
Online (OCS)	Gy (Diameter)	OCS	Prepaid; real-time credit control
Offline (OFCS)	Gz (GTP')	CDF/CGF	Postpaid; CDR generation

Charging triggers:

- Volume threshold (e.g., every 1 MB)
- Time threshold (e.g., every 60 seconds)
- QoS change, bearer modification
- Re-authorization from OCS

3.4 Packet Filtering and Routing

- **DL:** Internet → P-GW: Match destination IP to UE session, apply SDF filter to select bearer, encapsulate in GTP-U toward S-GW
- **UL:** UE → P-GW: Decapsulate GTP-U, apply NAT (if private addressing), route to Internet
- **NAT/NAPT:** P-GW often performs NAT for IPv4 (private UE addresses → public)
- **Header compression:** Not at P-GW (done at PDCP layer in RAN)

3.5 SGi Interface

- SGi connects P-GW to **external PDNs** (Internet, IMS, enterprise VPNs)
- IP-based interface (no GTP)
- Equivalent to Gi in 3G (GGSN → Internet)
- Multiple PDNs supported (each identified by APN)
- May connect to:

- Internet (public APN)
- IMS core (ims APN)
- Enterprise VPN (private APN)
- Content/CDN platforms

3.6 UL/DL TFT – Mapping Packets to Bearers

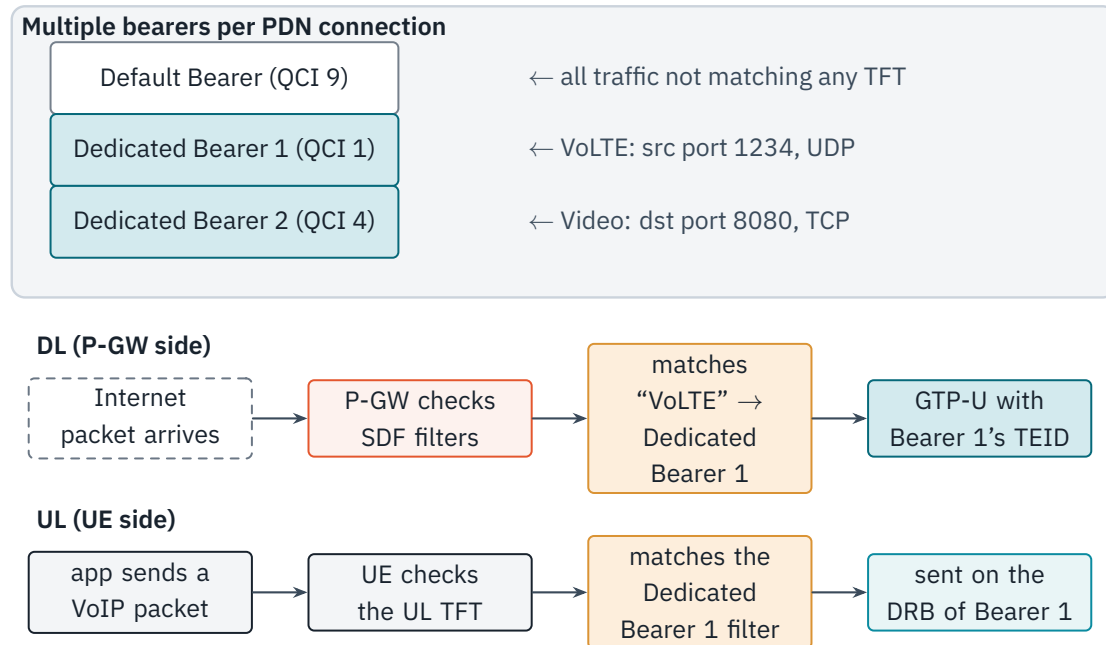


Figure 13.1 Several bearers per PDN connection and how packets are mapped to them

4 GTP Protocol

4.1 GTP-C (GTP Control Plane) – GTPv2-C

GTP-C manages tunnel/session lifecycle on S11, S5/S8, S10, S3 interfaces.

Key Messages:

Message	Direction	Purpose
Create Session Request/ Response	MME→S-GW→P-GW	Establish PDN connection + default bearer
Modify Bearer Request/ Response	MME→S-GW	Update tunnel endpoints (HO, service request)
Delete Session Request/ Response	MME→S-GW→P-GW	Tear down PDN connection
Create Bearer Request/Response	P-GW→S-GW→MME	Dedicated bearer activation
Update Bearer Request/ Response	P-GW→S-GW→MME	QoS modification
Delete Bearer Request/Response	P-GW→S-GW→MME	Dedicated bearer deactivation
Downlink Data Notification	S-GW→MME	Trigger paging for idle UE
Release Access Bearers	MME→S-GW	UE goes idle, release S1-U

GTPv2-C Header:

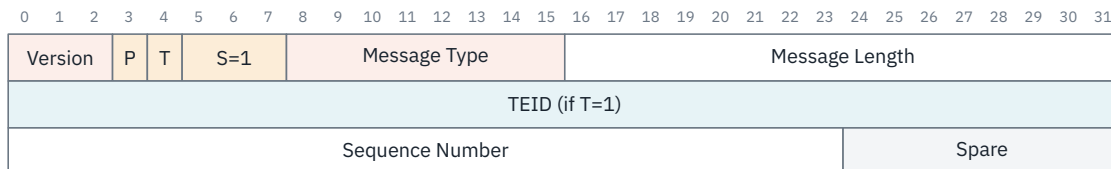


Figure 13.2 GTPv2-C header

TEID (Tunnel Endpoint Identifier):

- 32-bit identifier, locally unique per node
- Each side assigns its own TEID and communicates it to the peer
- Allows multiplexing multiple tunnels on same IP:port

4.2 GTP-U (GTP User Plane)

GTP-U encapsulates user IP packets for transport between EPC nodes.

GTP-U Header:

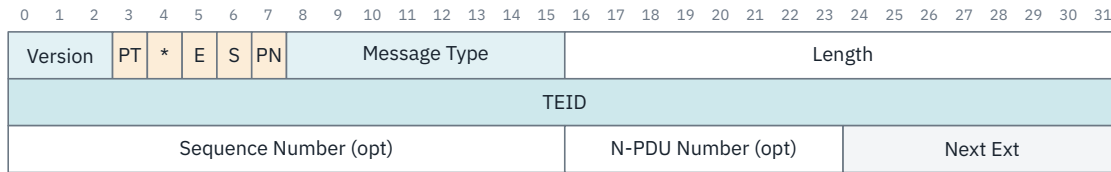


Figure 13.3 GTP-U header

- **Message Type 0xFF (255):** T-PDU — contains user data (most common)
- **Transport:** UDP port 2152
- **T-PDU:** The encapsulated user IP packet
- **Sequence Number:** Optional, used for reordering during path switch/handover

4.3 GTP-C vs GTP-U Comparison

Aspect	GTP-C (GTPv2-C)	GTP-U (GTPv1-U)
Purpose	Session/tunnel management	User data transport
Version	v2 (in LTE)	v1 (unchanged from 3G)
Transport	UDP port 2123	UDP port 2152
Reliability	Request/Response + retransmission	No ACK (relies on upper layers)
TEID	Identifies control session	Identifies user data tunnel (per-bearer)
Messages	Create/Modify/Delete Session, etc.	T-PDU (0xFF), Echo, Error Indication
Interfaces	S11, S5/S8, S10, S3	S1-U, S5/S8 (user plane)
Payload	IE (Information Elements) structured	Raw IP packet (T-PDU)
Header size	12 bytes (min)	8 bytes (min), 12 with options
Statefulness	Stateful (sessions)	Stateless (per-packet forwarding)

5 Packet Path Trace (End-to-End with Encapsulation)

5.1 Downlink: Internet → UE

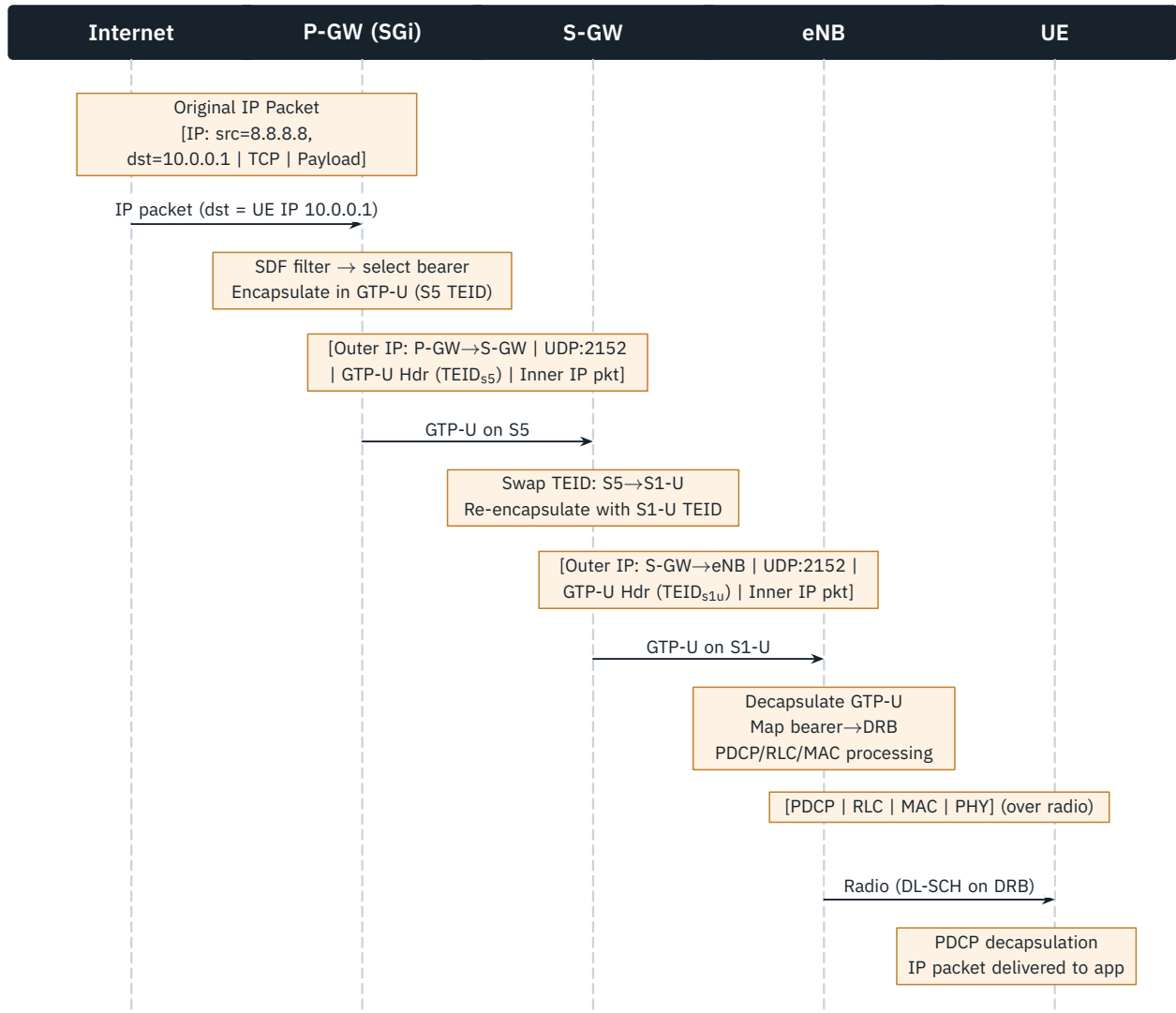


Figure 13.4 Downlink packet trace from the Internet to the UE with each encapsulation

5.2 Headers at Each Point

Segment 1: Internet → P-GW (SGi)

IP header src=8.8.8.8 dst=10.0.0.1	TCP	Application data
---	------------	------------------

Segment 2: P-GW → S-GW (S5/S8 GTP-U tunnel)

Outer IP src=P-GW IP dst=S-GW IP	UDP dst=2152	GTP-U header TEID=0xABCD Type=0xFF (TPDU)	Inner IP src=8.8.8.8 dst=10.0.0.1	TCP	Data
---	------------------------	--	--	------------	------

Segment 3: S-GW → eNB (S1-U GTP-U tunnel)

Outer IP src=S-GW IP dst=eNB IP	UDP dst=2152	GTP-U header TEID=0x1234 Type=0xFF (TPDU)	Inner IP src=8.8.8.8 dst=10.0.0.1	TCP	Data
--	------------------------	--	--	------------	------

Segment 4: eNB → UE (radio / air interface)

PHY	MAC	RLC	PDCP (ciphered)	Inner IP packet (original IP packet delivered to the UE)
------------	------------	------------	---------------------------	--

Figure 13.5 Headers of a downlink packet on each segment

5.3 GTP Tunnel Structure Diagram

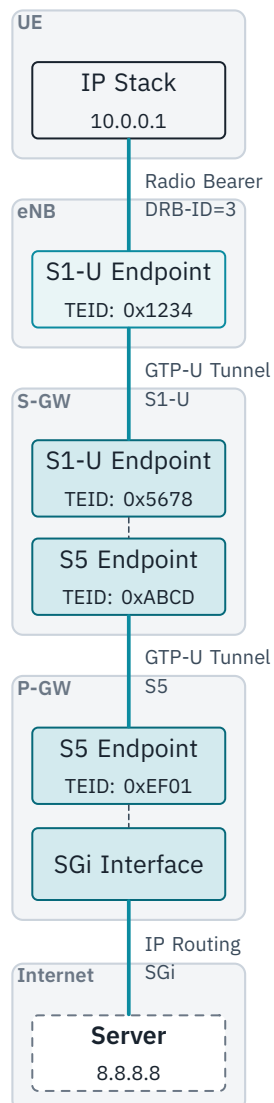


Figure 13.6 GTP tunnel endpoints and TEIDs along the user-plane path

Note: Each GTP-U tunnel segment has **two TEIDs** — one for each direction:

- S1-U UL: eNB→S-GW uses S-GW's TEID (0x5678)
- S1-U DL: S-GW→eNB uses eNB's TEID (0x1234)

6 CUPS: Control and User Plane Separation

6.1 Concept (3GPP Release 14+)

CUPS splits S-GW and P-GW each into control-plane and user-plane components:

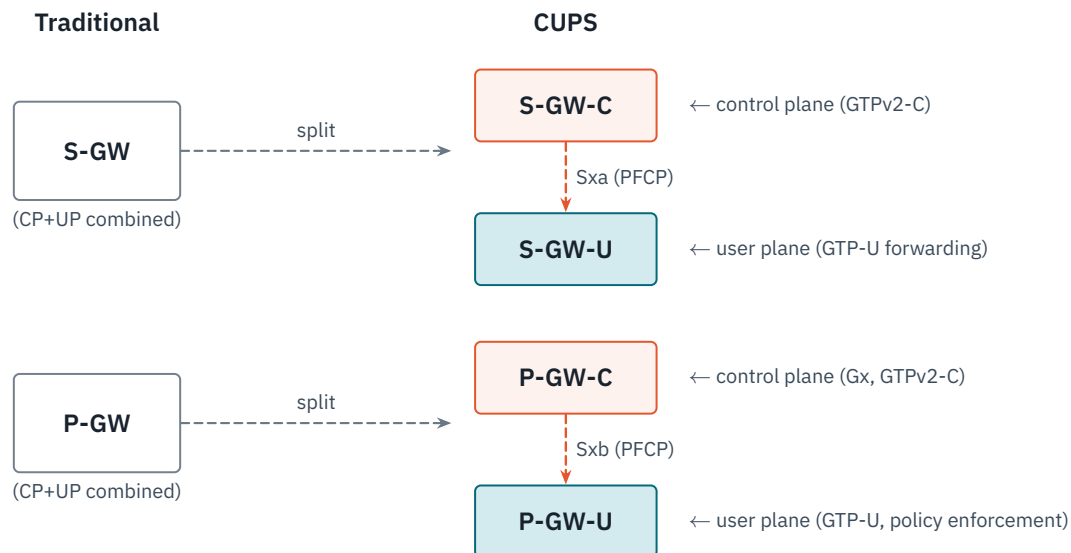


Figure 13.7 CUPS splits S-GW and P-GW into control-plane and user-plane parts

6.2 Sx Interface and PFCP Protocol

PFCP (Packet Forwarding Control Protocol) — 3GPP TS 29.244:

- Runs between CP and UP components (Sxa for S-GW, Sxb for P-GW, Sxc for TDF)
- **Session Establishment:** CP tells UP to create forwarding rules
- **Session Modification:** CP updates rules (e.g., new TEID after handover)
- **Session Deletion:** CP instructs UP to remove forwarding state

PFCP Key Concepts:

| Element | Purpose |

| --- | --- |

| PDR (Packet Detection Rule) | Match criteria (TEID, IP, port) |

| FAR (Forwarding Action Rule) | What to do (forward, buffer, drop) |

| QER (QoS Enforcement Rule) | Rate limiting (GBR/MBR) |

| URR (Usage Reporting Rule) | Volume/time measurement for charging |

| BAR (Buffering Action Rule) | Buffering behavior for idle UE |

6.3 Benefits of CUPS

1. **Distributed user plane:** Deploy S-GW-U / P-GW-U at edge (low latency for MEC)
2. **Centralized control:** Keep S-GW-C / P-GW-C in central DC (simpler management)
3. **Independent scaling:** Scale UP with traffic volume, CP with signaling load

4. **Flexible deployment:** Multiple UP instances per CP, or vice versa
5. **5G readiness:** CUPS architecture directly maps to 5G SMF (CP) + UPF (UP) split

6.4 CUPS Architecture Diagram

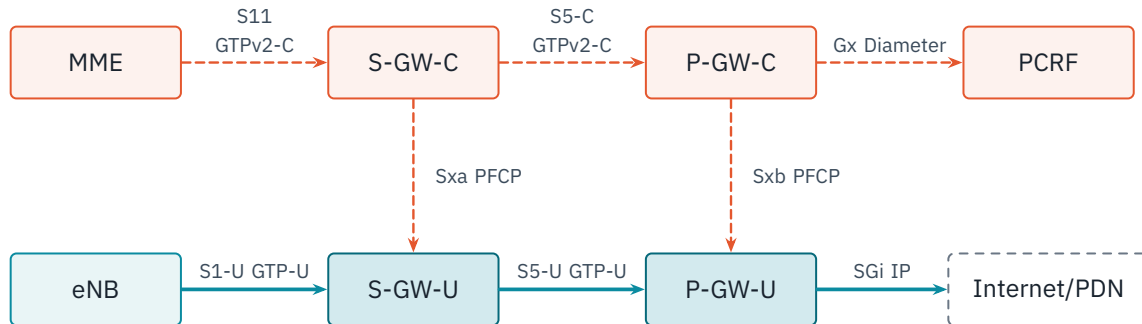


Figure 13.8 CUPS: S-GW and P-GW split into control and user planes

Blue = Control Plane | Green = User Plane

7 Evolution: S-GW + P-GW → UPF (5G)

7.1 Why Merged into UPF?

In 5G, S-GW and P-GW user-plane functions are **merged into a single UPF (User Plane Function)**:

Aspect	LTE (S-GW + P-GW)	5G (UPF)
Nodes	2 separate (S-GW, P-GW)	1 unified (UPF)
GTP hops	2 (S1-U + S5)	1 (N3 only, or N3+N9 for chaining)
Control	S-GW-C + P-GW-C	SMF (single CP for UP)
CP-UP protocol	Sxa/Sxb PFCP	N4 PFCP (same protocol, evolved)
Deployment	Typically centralized	Distributed (edge UPF, anchor UPF)
Mobility anchor	Fixed P-GW	Flexible (SSC modes 1/2/3)
Slicing	N/A	UPF per slice

7.2 Why the Merge Makes Sense

1. **Eliminates one GTP-U hop:** S1-U + S5 → just N3 (+ optional N9 between UPFs)
 - Lower latency, fewer encap/decap operations
2. **Simplified forwarding:** One node handles both local mobility and PDN anchoring

3. **Edge deployment:** UPF placed close to user for MEC/URLLC — no need for separate S-GW-U at edge
4. **Flexible anchoring (SSC modes):**
 - SSC 1: Fixed UPF (like LTE P-GW) — IP continuity
 - SSC 2: UPF can change — new IP address (for non-persistent sessions)
 - SSC 3: Multi-homing — old + new UPF active simultaneously during transition
5. **UPF chaining:** Multiple UPFs in series (I-UPF as intermediate, PSA-UPF as anchor) replaces S-GW→P-GW chain with more flexible topology

7.3 Interface Mapping

LTE	5G
S1-U (eNB ↔ S-GW)	N3 (gNB ↔ UPF)
S5/S8 (S-GW ↔ P-GW)	N9 (UPF ↔ UPF, if chained)
SGi (P-GW ↔ Internet)	N6 (UPF ↔ DN)
S11 (MME ↔ S-GW)	N4 (SMF ↔ UPF) [PFCP]
Gx (PCRF ↔ P-GW)	N7 (PCF ↔ SMF) [SBI]

Summary

Function	S-GW	P-GW
User-plane forwarding	✓	✓
GTP-U termination	S1-U + S5	S5 + SGi
IP address allocation	✗	✓
Policy enforcement (PCC)	✗	✓
Charging (online)	✗	✓ (Gy)
Charging (offline)	✓ (CDRs)	✓ (Gz)
DL buffering + paging trigger	✓	✗
Local mobility anchor	✓	✗
PDN/IP anchor	✗	✓
Lawful interception	✓	✓
NAT	✗	✓
TFT/SDF enforcement	✗	✓